LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 3 Test (Set B): Metals and Non-metals

Chapter / Topic: Metals and Non-metals

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Which non-metal is a liquid at room temperature? **[1]**

(a) Iodine (b) Mercury
(c) Bromine (d) Sulphur

Q2. The property of metals that allows them to be drawn into thin wires is: **[1]**

(a) malleability (b) sonority
(c) lustre (d) ductility

Q3. Which of these metals reacts only with steam, not with cold or hot water? **[1]**

(a) Sodium (b) Iron
(c) Potassium (d) Calcium

Q4. In electrolytic refining, the anode mud consists of: **[1]**

(a) soluble impurities (b) insoluble impurities
(c) the pure metal (d) the electrolyte

Q5. Which of the following oxides is basic? **[1]**

(a) SO_2 (b) CO_2
(c) SO_3 (d) Na_2O

Q6. **Assertion (A):** Ionic compounds are generally solids and somewhat hard.
Reason (R): There is a strong force of attraction between the positive and negative ions. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why is sodium kept immersed in kerosene oil? **[2]**

OR

What is anodising? Why is it done?

Q8. Name two metals that are the best conductors of heat, and two that are comparatively poor conductors of heat. **[2]**

SECTION C

6 marks

Q9. Element X has atomic number 12, and element Y has atomic number 17. **[3]**

(i) Write the electronic configuration of X and Y. **[1]**

(ii) Write the formula of the compound formed between X and Y. **[1]**

(iii) Name the type of bond, and give two properties of this compound. **[1]**

Q10. Describe an activity to compare the reactivity of magnesium, aluminium, zinc and iron with dilute hydrochloric acid. Write the equations for magnesium and aluminium. **[3]**

OR

Define the terms mineral, ore and gangue.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Explain how copper is obtained from copper sulphide (Cu_2S). Give balanced equations. **[3]**

(ii) Why can carbon not be used to reduce the oxides of sodium, magnesium and aluminium? How are these metals obtained? **[2]**

OR

(a) Give reasons: (i) Carbonate and sulphide ores are converted into oxides before reduction. (ii) Aluminium is very reactive, yet it is used to make cooking utensils. (iii) Gold and platinum are used to make jewellery. [3]
(b) What is solder? Why is it used for welding electrical wires? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. When iron(III) oxide is mixed with aluminium powder and ignited, the reaction gives out so much heat that the iron formed is molten. This molten iron is used to join railway tracks and cracked machine parts. This is called the thermite reaction. **[4]**

(i) Write the balanced equation for the thermite reaction. **[1]**

(ii) Which substance acts as the reducing agent? **[1]**

(iii) Why is aluminium able to reduce iron oxide? Name two types this reaction belongs to. **[2]**

~ All the best! ~